

MIA-2026-Project

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1 Ventilator Pressure Prediction

USC MIA-2026, XTAI Project inspired by Kaggle Competition :

<https://www.kaggle.com/competitions/ventilator-pressure-prediction>

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2 1. Introduction

2.1 1.1 Background

Mechanical ventilation is a critical medical intervention used when patients are unable to breathe adequately on their own. A ventilator delivers oxygen into the patient's lungs through a tube inserted into the windpipe while the patient is sedated.

Mechanical ventilation is a highly clinician-intensive process. This limitation became particularly visible during the early stages of the COVID-19 pandemic when the demand for ventilators and trained medical staff increased dramatically worldwide.

Developing and testing new ventilator control strategies is costly and complex, especially before reaching clinical trial stages. High-fidelity computational simulators provide a promising solution by enabling safe and cost-effective experimentation.

2.2 1.2 Problem Statement

Traditional ventilator control systems rely on **PID (Proportional–Integral–Derivative) controllers**.

Although PID controllers are widely used, they may struggle to adapt to patients with significantly different lung characteristics.

Existing ventilator simulators are often trained as **ensembles of models**, each corresponding to a specific lung configuration. However, lung mechanics vary continuously across patients.

Therefore, a more general **parametric modeling approach** is required.

2.3 1.3 Research Objective

The objective of this project is to develop a **data-driven simulator** capable of predicting airway pressure dynamics during mechanical ventilation.

The model must account for:

- **Control inputs** from ventilator valves
- **Lung mechanical properties**
- **Temporal airflow dynamics**

The central hypothesis is that:

Deep learning models can better generalize across lungs with varying resistance and compliance compared to traditional PID controllers.

3 2. Ventilator System Overview

3.1 2.1 Experimental Setup

The dataset was generated using a **modified open-source ventilator** connected to an artificial **bellows test lung**.

The ventilator system includes:

- **Two control signals**
- **One target variable (airway pressure)**

The ventilator manipulates airflow using inspiratory and expiratory valves.

3.2 2.2 Control Inputs

3.2.1 Inspiratory Valve (u_{in})

Continuous variable ranging from **0 to 100** representing the percentage opening of the inspiratory valve.

Value	Meaning
0	Valve closed
100	Valve fully open

3.2.2 Expiratory Valve (u_{out})

Binary variable indicating whether the expiratory valve is open.

Value	Meaning
0	Valve closed
1	Valve open

3.3 2.3 Target Variable

3.3.1 Airway Pressure (**pressure**)

Measured airway pressure inside the respiratory circuit.

Units: **cmH O**

The goal of the model is to predict this pressure at every timestep.

4 3. Dataset Description

4.1 3.1 Time Series Structure

Each breathing cycle corresponds to **one time series sequence**.

Dataset characteristics:

Property	Value
Sequence length	80 timesteps
Duration	~3 seconds
Target	airway pressure
Inputs	valve controls + lung parameters

Each sequence corresponds to **one breath** identified by **breath_id**.

4.2 3.2 Dataset Columns

4.2.1 Identifiers

Feature	Description
id	unique identifier for each timestep
breath_id	identifier grouping timesteps within the same breath

4.2.2 Lung Mechanical Properties

Feature	Meaning
R	lung resistance
C	lung compliance

These parameters define the physical behavior of the lung.

Higher resistance makes airflow more difficult, while higher compliance allows the lung to expand more easily.

4.2.3 Control Signals

Feature	Meaning
u_in	inspiratory valve opening
u_out	expiratory valve state

4.2.4 Time Variable

Feature	Meaning
time_step	timestamp within the breath

5 4. Ventilator Physics

The pressure dynamics inside the lung can be approximated by the equation:

$$P(t) = R \cdot F(t) + \frac{V(t)}{C}$$

where:

Symbol	Meaning
(P(t))	airway pressure
(F(t))	airflow
(V(t))	lung volume
(R)	lung resistance
(C)	lung compliance

The dataset provides airflow and valve signals, and the goal is to learn a model that approximates this underlying physiological relationship.

6 5. Time Series Modeling

6.1 5.1 Time Series Characteristics

A time series is a sequence of observations ordered in time:

$$x_1, x_2, x_3, \dots, x_T$$

Time series differ from traditional tabular data because observations exhibit **temporal dependencies**.

Key characteristics include:

- Autocorrelation
 - Sequential structure
 - Dynamic evolution
-

6.2 5.2 Time Series in Ventilator Data

In this dataset:

- Each breath forms a **short multivariate time series**
- Sequences are independent across breaths
- Within each breath, pressure evolves dynamically

Pressure depends on:

- Previous pressure values
- Airflow inputs
- Lung mechanical properties

Therefore, the ventilator system behaves as a **nonlinear dynamical system**.

7 6. Feature Engineering

Feature engineering aims to provide the model with **temporal context and physical signals** describing airflow dynamics.

The engineered features capture:

- airflow accumulation
- breathing phase transitions
- past and future airflow patterns

These signals help the model approximate the **underlying respiratory mechanics**.

(your feature engineering section can follow here)

8 7. Model Architecture

Sequential models are particularly suitable for this problem because they capture **temporal dependencies**.

Common architectures for this task include:

- Recurrent Neural Networks (RNN)
- Long Short-Term Memory (LSTM)
- Gated Recurrent Units (GRU)
- Transformer architectures

These models can learn nonlinear relationships between airflow control signals and airway pressure.

9 8. Evaluation Metric

For our model the evaluation metric is MAE. Using MAE as the training loss means that the model optimizes exactly what will be evaluated.

The **Mean Absolute Error (MAE)** measures the average absolute difference between the predicted value and the true value.

$$\text{MAE} = \frac{1}{N} \sum |y_i - \hat{y}_i|$$

Where:

y = true value

$\hat{y}$ = predicted value

n = number of samples

Why do we use MAE?

Minimizing MAE makes the model estimate the conditional median of the target distribution.

This is beneficial when: - target distribution is skewed
- there are outliers

Ventilator pressure distributions often have:

- plateaus
- jumps
- asymmetric noise

So predicting the median behavior is reasonable.

MAE produces interpretable errors.

Since MAE is measured in the same units as the target, the interpretation of the results is very intuitive for Clinicians and model evaluation.

Example: MAE = 0.5

Means on average the prediction is off by 0.5 pressure units.

MAE penalizes errors linearly.

Because MAE does not square the error, it is less sensitive to extreme values.

Example : True pressure value = 20

Prediction	Error	MAE	MSE
21	1	1	1
22	2	2	4
40	20	20	400

With MSE:

one bad prediction (40) would dominate the loss.

With MAE:

errors remain proportional.

This is useful when sensor noise or rare events exist.

The gradient magnitude is constant

Large errors do not explode gradients.

This helps when training deep sequence models such as LSTMs or Transformers.

10 9. Model Interpretability

To understand model behavior, we apply **SHAP (SHapley Additive Explanations)**.

SHAP assigns each feature a contribution value explaining its impact on a prediction.

A prediction can be decomposed as:

$$f(x) = \phi_0 + \sum_{i=1}^d \phi_i$$

where (ϕ_i) represents the contribution of feature (i).

This allows analysis of:

- global feature importance
- local explanations for individual breaths
- temporal importance across timesteps

11 10. Conclusion

This project formulates ventilator simulation as a **multivariate time-series prediction problem**.

By combining:

- physics-inspired features
- sequential deep learning models
- interpretability techniques

the system aims to accurately model airway pressure dynamics across different lung configurations. Such models may support the development of **adaptive ventilator control systems** capable of improving respiratory care.

12 IMPORTS

```
[ ]: import numpy as np # linear algebra
import pandas as pd # data processing, CSV file I/O (e.g. pd.read_csv)
import time
import warnings
import gc
import multiprocessing
import os, random, math, psutil, pickle
from datetime import datetime
import matplotlib.pyplot as plt
import seaborn as sns
from datetime import datetime

#~~~~~
#                               sklearn
#~~~~~
from sklearn.preprocessing import StandardScaler
from sklearn.preprocessing import LabelEncoder
from sklearn.preprocessing import RobustScaler, normalize
from sklearn.model_selection import train_test_split, GroupKFold, KFold
import tensorflow as tf
! conda env export > environment.yml

#~~~~~
#                               Keras
#~~~~~
import tensorflow.keras.backend as K
from tensorflow.keras.losses import MeanSquaredError as mse_loss
from tensorflow.keras.utils import *
from tensorflow.keras import Sequential
from tensorflow.keras.layers import *
from tensorflow.keras.callbacks import *
from tensorflow.keras.models import Model, Sequential, model_from_json
from tensorflow.keras import regularizers

from tensorflow.keras.layers import Dense, LSTM, Bidirectional
import tensorflow as tf
from tensorflow.keras.callbacks import ModelCheckpoint

import shap
```

```
import numpy as np
```

2026-03-11 10:33:55.037102: I tensorflow/core/platform/cpu_feature_guard.cc:210] This TensorFlow binary is optimized to use available CPU instructions in performance-critical operations.

To enable the following instructions: AVX2 FMA, in other operations, rebuild TensorFlow with the appropriate compiler flags.

/bin/bash: /home/gpaolo/miniconda3/lib/libtinfo.so.6: no version information available (required by /bin/bash)

/home/gpaolo/miniconda3/envs/dl_env/lib/python3.12/site-packages/tqdm/auto.py:21: TqdmWarning: IPProgress not found. Please update jupyter and ipywidgets. See https://ipywidgets.readthedocs.io/en/stable/user_install.html
from .autonotebook import tqdm as notebook_tqdm

13 Set Notebook Options

```
[2]: print("Tensorflow version " + tf.__version__)  
print("Num GPUs Available: ", len(tf.config.list_physical_devices('GPU')))  
warnings.filterwarnings('ignore')  
pd.set_option('display.max_columns', None)  
pd.set_option('display.max_rows', None)  
pd.set_option('display.float_format', lambda x: '%.5f' % x)
```

Tensorflow version 2.20.0
Num GPUs Available: 1

14 Functions

14.1 ADD FEATURES

This function creates engineered features for the ventilator time-series dataset, where each `breath_id` represents a single breathing cycle and each row corresponds to a time step within that breath.

The dataset represents ventilator breathing cycles. Each `breath_id` corresponds to a complete breath and contains multiple time steps describing airflow and valve behavior.

The goal of feature engineering is to provide the model with information about:

- Airflow dynamics
- Accumulated lung volume
- Breathing phase transitions
- Physical lung properties

These signals help the model predict airway pressure at each time step.

Since the dataset provides airflow (`u_in`) and valve state (`u_out`), we engineer features that approximate:

- Flow history
- Accumulated volume
- Breathing phase
- Lung mechanical properties

– Volume / Accumulation Features

These features approximate the amount of air delivered to the lungs over time. Pressure increases as lung volume increases, so cumulative airflow helps estimate how full the lungs are during a breath.

Feature	Meaning
<code> u_in </code>	airflow entering the lungs
<code> time_step </code>	time

Meaning

This approximates the integral of airflow over time.

It estimates the volume of air delivered during the breath so far.

This is important because pressure depends on lung volume.

Cumulative air-flow

Pressure in lungs increases as volume increases.

Feature | Meaning

<code> u_in_cumsum </code>	Running total of airflow entering during the breath.
----------------------------	--

Lag Features (Past Information)

Pressure depends on previous airflow because lungs have memory.

Feature | Meaning

<code> u_in_lag1 </code>	airflow at previous timestep
<code> u_out_lag1 </code>	valve state at previous timestep
<code> u_in_lag2 </code>	airflow 2 timesteps ago
<code> u_in_lag3 </code>	airflow 3 timesteps ago
<code> u_in_lag4 </code>	airflow 4 timesteps ago

Forward Lag (Future Information) For models trained on entire sequences, future values help capture patterns in the breathing curve. A sudden drop in airflow indicates exhalation phase starting.

Feature	Meaning
<code>u_in_lag_back1</code>	Airflow at the next timestep
<code>u_in_lag_back2</code>	Airflow two timesteps ahead
<code>u_in_lag_back3</code>	Airflow three timesteps ahead
<code>u_in_lag_back4</code>	Airflow four timesteps ahead

Past Valve State (u_out) The variable `u_out` indicates whether the exhalation valve is open. Past valve states help detect the breathing phase transition between inhalation and exhalation. Past and future exhalation valve states.

Feature	Meaning
<code>u_out_lag1</code>	Valve state 1 timestep before
<code>u_out_lag2</code>	Valve state 2 timesteps before
<code>u_out_lag3</code>	Valve state 3 timesteps before
<code>u_out_lag4</code>	state 4 timesteps before

Future Valve State

Future valve information helps anticipate when exhalation will begin, which typically causes a rapid pressure decrease. |Feature | Meaning

u_out_lag_back1	Valve state next timestep
u_out_lag_back2	Valve state two timesteps ahead
u_out_lag_back3	Valve state three timesteps ahead
u_out_lag_back4	Valve state four timesteps ahead

Lung Mechanical Properties These parameters define the physical properties of the lung and strongly influence pressure behavior.

Feature	Meaning
R	Lung resistance affecting pressure from airflow
C	Lung compliance affecting pressure from lung volume
R__C	Combined lung type formed from R and C

```
[3]: ## Functions
#from google.colab import drive
#drive.mount('/content/gdrive', force_remount=True)
#!ls "/content/gdrive/My Drive/kaggle"
def add_features(df):#022
    df['area'] = df['time_step'] * df['u_in']
    df['area'] = df.groupby('breath_id')['area'].cumsum()
    df['u_in_cumsum'] = (df['u_in']).groupby(df['breath_id']).cumsum()
    df['u_in_lag1'] = df.groupby('breath_id')['u_in'].shift(1)
    df['u_out_lag1'] = df.groupby('breath_id')['u_out'].shift(1)
    df['u_in_lag_back1'] = df.groupby('breath_id')['u_in'].shift(-1)
    df['u_out_lag_back1'] = df.groupby('breath_id')['u_out'].shift(-1)
    df['u_in_lag2'] = df.groupby('breath_id')['u_in'].shift(2)
    df['u_out_lag2'] = df.groupby('breath_id')['u_out'].shift(2)
    df['u_in_lag_back2'] = df.groupby('breath_id')['u_in'].shift(-2)
    df['u_out_lag_back2'] = df.groupby('breath_id')['u_out'].shift(-2)
    df['u_in_lag3'] = df.groupby('breath_id')['u_in'].shift(3)
    df['u_out_lag3'] = df.groupby('breath_id')['u_out'].shift(3)
    df['u_in_lag_back3'] = df.groupby('breath_id')['u_in'].shift(-3)
    df['u_out_lag_back3'] = df.groupby('breath_id')['u_out'].shift(-3)
    df['u_in_lag4'] = df.groupby('breath_id')['u_in'].shift(4)
    df['u_out_lag4'] = df.groupby('breath_id')['u_out'].shift(4)
    df['u_in_lag_back4'] = df.groupby('breath_id')['u_in'].shift(-4)
    df['u_out_lag_back4'] = df.groupby('breath_id')['u_out'].shift(-4)
    df = df.fillna(0)

    df['R'] = df['R'].astype(str)
    df['C'] = df['C'].astype(str)
    df['R__C'] = df["R"].astype(str) + '__' + df["C"].astype(str)
    df = pd.get_dummies(df)
    return df
```

15 Load data files

```
[ ]: #~~~~~  
# Load data files  
#~~~~~  
path = "./data/"  
start_time = time.time()  
#dataset = pd.read_csv(path + 'train.csv')  
dataset = pd.read_csv(path + 'train.csv.zip')  
print('Load files took %s seconds' % (time.time() - start_time))  
  
# print datasets info  
display(dataset.info())
```

Load files took 4.275898456573486 seconds

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 6036000 entries, 0 to 6035999

Data columns (total 8 columns):

#	Column	Dtype
0	id	int64
1	breath_id	int64
2	R	int64
3	C	int64
4	time_step	float64
5	u_in	float64
6	u_out	int64
7	pressure	float64

dtypes: float64(3), int64(5)

memory usage: 368.4 MB

None

16 Check Missing values

```
[6]: # Check Missing values  
dataset.isnull().sum()
```

```
[6]: id          0  
breath_id      0  
R              0  
C              0  
time_step      0  
u_in           0  
u_out          0  
pressure       0  
dtype: int64
```

17 Data Pre-processing

```
[7]: dataset = add_features(dataset)
print(f'Dataset features {dataset.shape[1]}')
dataset.head(10)
```

Dataset features 39

```
[7]:
```

	id	breath_id	time_step	u_in	u_out	pressure	area	u_in_cumsum	\
0	1	1	0.00000	0.08333	0	5.83749	0.00000	0.08333	
1	2	1	0.03365	18.38304	0	5.90779	0.61863	18.46638	
2	3	1	0.06751	22.50928	0	7.87625	2.13833	40.97565	
3	4	1	0.10154	22.80882	0	11.74287	4.45439	63.78448	
4	5	1	0.13576	25.35585	0	12.23499	7.89659	89.14033	
5	6	1	0.16970	27.25987	0	12.86771	12.52253	116.40019	
6	7	1	0.20371	27.12749	0	14.69556	18.04861	143.52768	
7	8	1	0.23772	26.80773	0	15.89070	24.42142	170.33541	
8	9	1	0.27178	27.86472	0	15.53919	31.99439	198.20012	
9	10	1	0.30573	28.31304	0	15.75009	40.65058	226.51316	

	u_in_lag1	u_out_lag1	u_in_lag_back1	u_out_lag_back1	u_in_lag2	\
0	0.00000	0.00000	18.38304	0.00000	0.00000	
1	0.08333	0.00000	22.50928	0.00000	0.00000	
2	18.38304	0.00000	22.80882	0.00000	0.08333	
3	22.50928	0.00000	25.35585	0.00000	18.38304	
4	22.80882	0.00000	27.25987	0.00000	22.50928	
5	25.35585	0.00000	27.12749	0.00000	22.80882	
6	27.25987	0.00000	26.80773	0.00000	25.35585	
7	27.12749	0.00000	27.86472	0.00000	27.25987	
8	26.80773	0.00000	28.31304	0.00000	27.12749	
9	27.86472	0.00000	26.86676	0.00000	26.80773	

	u_out_lag2	u_in_lag_back2	u_out_lag_back2	u_in_lag3	u_out_lag3	\
0	0.00000	22.50928	0.00000	0.00000	0.00000	
1	0.00000	22.80882	0.00000	0.00000	0.00000	
2	0.00000	25.35585	0.00000	0.00000	0.00000	
3	0.00000	27.25987	0.00000	0.08333	0.00000	
4	0.00000	27.12749	0.00000	18.38304	0.00000	
5	0.00000	26.80773	0.00000	22.50928	0.00000	
6	0.00000	27.86472	0.00000	22.80882	0.00000	
7	0.00000	28.31304	0.00000	25.35585	0.00000	
8	0.00000	26.86676	0.00000	27.25987	0.00000	
9	0.00000	26.76280	0.00000	27.12749	0.00000	

	u_in_lag_back3	u_out_lag_back3	u_in_lag4	u_out_lag4	u_in_lag_back4	\
0	22.80882	0.00000	0.00000	0.00000	25.35585	
1	25.35585	0.00000	0.00000	0.00000	27.25987	

2	27.25987	0.00000	0.00000	0.00000	27.12749
3	27.12749	0.00000	0.00000	0.00000	26.80773
4	26.80773	0.00000	0.08333	0.00000	27.86472
5	27.86472	0.00000	18.38304	0.00000	28.31304
6	28.31304	0.00000	22.50928	0.00000	26.86676
7	26.86676	0.00000	22.80882	0.00000	26.76280
8	26.76280	0.00000	25.35585	0.00000	27.99327
9	27.99327	0.00000	27.25987	0.00000	26.78990

	u_out_lag_back4	R_20	R_5	R_50	C_10	C_20	C_50	R__C_20__10	\
0	0.00000	True	False	False	False	False	True	False	
1	0.00000	True	False	False	False	False	True	False	
2	0.00000	True	False	False	False	False	True	False	
3	0.00000	True	False	False	False	False	True	False	
4	0.00000	True	False	False	False	False	True	False	
5	0.00000	True	False	False	False	False	True	False	
6	0.00000	True	False	False	False	False	True	False	
7	0.00000	True	False	False	False	False	True	False	
8	0.00000	True	False	False	False	False	True	False	
9	0.00000	True	False	False	False	False	True	False	

	R__C_20__20	R__C_20__50	R__C_50__10	R__C_50__20	R__C_50__50	\
0	False	True	False	False	False	
1	False	True	False	False	False	
2	False	True	False	False	False	
3	False	True	False	False	False	
4	False	True	False	False	False	
5	False	True	False	False	False	
6	False	True	False	False	False	
7	False	True	False	False	False	
8	False	True	False	False	False	
9	False	True	False	False	False	

	R__C_5__10	R__C_5__20	R__C_5__50
0	False	False	False
1	False	False	False
2	False	False	False
3	False	False	False
4	False	False	False
5	False	False	False
6	False	False	False
7	False	False	False
8	False	False	False
9	False	False	False

18 TARGET EXTRACTION

```
[ ]: # -----  
# Reshape target to 3D array for LSTM  
# -----  
# Ensure sorted  
dataset = dataset.sort_values(["breath_id", "time_step"])  
targets = dataset['pressure'].values.reshape(-1, 80)  
n = targets.shape[0]
```

19 Dataset Split train,val,test

```
[9]: # Correct Sequence-Preserving Split  
# train 50% of the data, validate 10%, and test on the remaining 40%  
# Entire breaths stay together  
# No timestep from one breath appears in multiple splits  
# No leakage across sequences  
# Proper LSTM training  
  
# Get unique sequences  
breaths = dataset["breath_id"].unique()  
n = len(breaths)  
  
train_end = int(0.5 * n)  
val_end   = int(0.6 * n)  
  
train_ids = breaths[:train_end]  
val_ids   = breaths[train_end:val_end]  
test_ids  = breaths[val_end:]  
  
train = dataset[dataset["breath_id"].isin(train_ids)].copy()  
val    = dataset[dataset["breath_id"].isin(val_ids)].copy()  
test   = dataset[dataset["breath_id"].isin(test_ids)].copy()  
  
print(train["breath_id"].nunique(),  
      val["breath_id"].nunique(),  
      test["breath_id"].nunique())  
  
# Extra Safety Check  
assert set(train["breath_id"]).isdisjoint(val["breath_id"])  
assert set(train["breath_id"]).isdisjoint(test["breath_id"])  
assert set(val["breath_id"]).isdisjoint(test["breath_id"])  
  
print("No sequence leakage ")
```

```
print("Train shape:", train.shape)
print("Val shape:", val.shape)
print("Test shape:", test.shape)
```

```
37725 7545 30180
No sequence leakage
Train shape: (3018000, 39)
Val shape: (603600, 39)
Test shape: (2414400, 39)
```

20 Make targets

```
[10]: y_train = train['pressure'].values.reshape(-1,80)
      y_val   = val['pressure'].values.reshape(-1,80)
      y_test  = test['pressure'].values.reshape(-1,80)

      print("Train Targets shape:", y_train.shape)
      print("Val Targets shape:", y_val.shape)
      print("Test Targets shape:", y_test.shape)

      #=====
      # Free memory
      #=====
      del targets
      gc.collect()
```

```
Train Targets shape: (37725, 80)
Val Targets shape: (7545, 80)
Test Targets shape: (30180, 80)
```

```
[10]: 20
```

21 Filter columns for training

```
[11]: #=====
      # Filter columns for training
      #=====

      drop_cols = ['pressure', 'id', 'breath_id', 'time_step']

      train.drop(drop_cols, axis=1, inplace=True)
      val.drop(drop_cols, axis=1, inplace=True)
      test.drop(drop_cols, axis=1, inplace=True)

      features = list(train.columns)
      print('Number of feature columns =', len(features))
```

```
#=====
# Free memory
#=====
del dataset
gc.collect()
```

Number of feature columns = 35

[11]: 0

21.1 SCALING

```
[12]: # -----
# SCALING
# -----
RS = RobustScaler()
train = RS.fit_transform(train)
test  = RS.transform(test)
val   = RS.transform(val)
```

21.2 RESHAPE train,val,test to sequences

```
[13]: # reshape train and test to 3D array for LSTM
# -----
# RESHAPE TO SEQUENCES
# -----
train = train.reshape(-1, 80, train.shape[-1])
val   = val.reshape(-1, 80, val.shape[-1])
test  = test.reshape(-1, 80, test.shape[-1])

print("Train shape:", train.shape)
print("Val shape:", val.shape)
print("Test shape:", test.shape)
```

Train shape: (37725, 80, 35)

Val shape: (7545, 80, 35)

Test shape: (30180, 80, 35)

```
[14]: print("Train Targets shape:", y_train.shape)
print("Val Targets shape:", y_val.shape)
print("Test Targets shape:", y_test.shape)
```

Train Targets shape: (37725, 80)

Val Targets shape: (7545, 80)

Test Targets shape: (30180, 80)

22 MODEL

```
[15]: # -----  
# MODEL  
# -----  
input_shape = train.shape[2]  
model = Sequential()  
  
model.add(Bidirectional(  
    LSTM(80, return_sequences=True),  
    input_shape=(80, input_shape)  
))  
  
model.add(Bidirectional(LSTM(80, return_sequences=True)))  
model.add(Bidirectional(LSTM(80, return_sequences=True)))  
  
model.add(TimeDistributed(Dense(80)))  
model.add(LeakyReLU(negative_slope=0.1))  
  
model.add(TimeDistributed(Dense(80)))  
model.add(LeakyReLU(negative_slope=0.1))  
  
model.add(TimeDistributed(Dense(1)))  
  
model.add(Flatten())  
  
# -----  
# COMPILE nd set optimizer  
# -----  
batch_size = 128  
epochs = 500  
steps_per_epoch = len(train) // batch_size  
total_steps = steps_per_epoch * epochs  
  
lr_schedule = tf.keras.optimizers.schedules.CosineDecay(  
    initial_learning_rate=3e-4,  
    decay_steps=total_steps,  
    alpha=0.01  
)  
  
optimizer = tf.keras.optimizers.AdamW(  
    learning_rate=lr_schedule,  
    weight_decay=1e-4  
)  
  
model.compile(  
    optimizer=optimizer,
```

```

        loss='mae',
        metrics=['mae']
    )

model.summary()

# -----
# CALLBACKS
# -----
early_stop = EarlyStopping(
    monitor='val_loss',
    patience=5,
    restore_best_weights=True
)

# -----
# Train
# -----
history = model.fit(
    train,
    y_train,
    validation_data=(val, y_val),
    epochs=epochs,
    batch_size=batch_size,
    callbacks=[early_stop],
    verbose=1
)

# -----
# Save the model
# -----
model.save('/ventilator.keras')

```

WARNING: All log messages before absl::InitializeLog() is called are written to STDERR

I0000 00:00:1773221670.257430 3337 gpu_device.cc:2020] Created device /job:localhost/replica:0/task:0/device:GPU:0 with 5580 MB memory: -> device: 0, name: NVIDIA GeForce RTX 3070 Laptop GPU, pci bus id: 0000:01:00.0, compute capability: 8.6

Model: "sequential"

Layer (type)	Output Shape	Param #
bidirectional (Bidirectional)	(None , 80, 160)	74,240

bidirectional_1 (Bidirectional)	(None, 80, 160)	154,240
bidirectional_2 (Bidirectional)	(None, 80, 160)	154,240
time_distributed (TimeDistributed)	(None, 80, 80)	12,880
leaky_re_lu (LeakyReLU)	(None, 80, 80)	0
time_distributed_1 (TimeDistributed)	(None, 80, 80)	6,480
leaky_re_lu_1 (LeakyReLU)	(None, 80, 80)	0
time_distributed_2 (TimeDistributed)	(None, 80, 1)	81
flatten (Flatten)	(None, 80)	0

Total params: 402,161 (1.53 MB)

Trainable params: 402,161 (1.53 MB)

Non-trainable params: 0 (0.00 B)

Epoch 1/500

2026-03-11 10:34:41.788298: I
external/local_xla/xla/stream_executor/cuda/cuda_dnn.cc:473] Loaded cuDNN
version 91900

295/295 39s 99ms/step -
loss: 2.4386 - mae: 2.4386 - val_loss: 1.0421 - val_mae: 1.0421

Epoch 2/500

295/295 26s 87ms/step -
loss: 0.9434 - mae: 0.9434 - val_loss: 0.8675 - val_mae: 0.8675

Epoch 3/500

295/295 27s 90ms/step -
loss: 0.8270 - mae: 0.8270 - val_loss: 0.8120 - val_mae: 0.8120

Epoch 4/500

295/295 27s 91ms/step -
loss: 0.7162 - mae: 0.7162 - val_loss: 0.6676 - val_mae: 0.6676

Epoch 5/500

295/295 27s 91ms/step -
loss: 0.6572 - mae: 0.6572 - val_loss: 0.6293 - val_mae: 0.6293

Epoch 6/500

295/295 27s 91ms/step -
 loss: 0.6336 - mae: 0.6336 - val_loss: 0.6217 - val_mae: 0.6217
 Epoch 7/500
 295/295 27s 93ms/step -
 loss: 0.6056 - mae: 0.6056 - val_loss: 0.5887 - val_mae: 0.5887
 Epoch 8/500
 295/295 28s 96ms/step -
 loss: 0.5812 - mae: 0.5812 - val_loss: 0.6116 - val_mae: 0.6116
 Epoch 9/500
 295/295 28s 95ms/step -
 loss: 0.5678 - mae: 0.5678 - val_loss: 0.5433 - val_mae: 0.5433
 Epoch 10/500
 295/295 29s 99ms/step -
 loss: 0.5487 - mae: 0.5487 - val_loss: 0.5410 - val_mae: 0.5410
 Epoch 11/500
 295/295 29s 97ms/step -
 loss: 0.5386 - mae: 0.5386 - val_loss: 0.5280 - val_mae: 0.5280
 Epoch 12/500
 295/295 29s 97ms/step -
 loss: 0.5269 - mae: 0.5269 - val_loss: 0.5292 - val_mae: 0.5292
 Epoch 13/500
 295/295 29s 100ms/step -
 loss: 0.5224 - mae: 0.5224 - val_loss: 0.5308 - val_mae: 0.5308
 Epoch 14/500
 295/295 29s 99ms/step -
 loss: 0.5128 - mae: 0.5128 - val_loss: 0.5147 - val_mae: 0.5147
 Epoch 15/500
 295/295 29s 99ms/step -
 loss: 0.5026 - mae: 0.5026 - val_loss: 0.4882 - val_mae: 0.4882
 Epoch 16/500
 295/295 29s 100ms/step -
 loss: 0.4991 - mae: 0.4991 - val_loss: 0.4846 - val_mae: 0.4846
 Epoch 17/500
 295/295 29s 97ms/step -
 loss: 0.4916 - mae: 0.4916 - val_loss: 0.5044 - val_mae: 0.5044
 Epoch 18/500
 295/295 30s 100ms/step -
 loss: 0.4905 - mae: 0.4905 - val_loss: 0.4796 - val_mae: 0.4796
 Epoch 19/500
 295/295 29s 99ms/step -
 loss: 0.4900 - mae: 0.4900 - val_loss: 0.4794 - val_mae: 0.4794
 Epoch 20/500
 295/295 28s 95ms/step -
 loss: 0.4769 - mae: 0.4769 - val_loss: 0.4791 - val_mae: 0.4791
 Epoch 21/500
 295/295 29s 99ms/step -
 loss: 0.4768 - mae: 0.4768 - val_loss: 0.4736 - val_mae: 0.4736
 Epoch 22/500

295/295 30s 102ms/step -
 loss: 0.4680 - mae: 0.4680 - val_loss: 0.4681 - val_mae: 0.4681
 Epoch 23/500
 295/295 29s 99ms/step -
 loss: 0.4681 - mae: 0.4681 - val_loss: 0.5128 - val_mae: 0.5128
 Epoch 24/500
 295/295 32s 107ms/step -
 loss: 0.4680 - mae: 0.4680 - val_loss: 0.4663 - val_mae: 0.4663
 Epoch 25/500
 295/295 30s 102ms/step -
 loss: 0.4598 - mae: 0.4598 - val_loss: 0.4560 - val_mae: 0.4560
 Epoch 26/500
 295/295 30s 102ms/step -
 loss: 0.4665 - mae: 0.4665 - val_loss: 0.5390 - val_mae: 0.5390
 Epoch 27/500
 295/295 32s 108ms/step -
 loss: 0.4591 - mae: 0.4591 - val_loss: 0.4549 - val_mae: 0.4549
 Epoch 28/500
 295/295 31s 106ms/step -
 loss: 0.4587 - mae: 0.4587 - val_loss: 0.4512 - val_mae: 0.4512
 Epoch 29/500
 295/295 31s 105ms/step -
 loss: 0.4473 - mae: 0.4473 - val_loss: 0.4432 - val_mae: 0.4432
 Epoch 30/500
 295/295 32s 107ms/step -
 loss: 0.4487 - mae: 0.4487 - val_loss: 0.4560 - val_mae: 0.4560
 Epoch 31/500
 295/295 32s 110ms/step -
 loss: 0.4496 - mae: 0.4496 - val_loss: 0.4505 - val_mae: 0.4505
 Epoch 32/500
 295/295 32s 109ms/step -
 loss: 0.4436 - mae: 0.4436 - val_loss: 0.4485 - val_mae: 0.4485
 Epoch 33/500
 295/295 33s 111ms/step -
 loss: 0.4420 - mae: 0.4420 - val_loss: 0.4473 - val_mae: 0.4473
 Epoch 34/500
 295/295 32s 107ms/step -
 loss: 0.4354 - mae: 0.4354 - val_loss: 0.4387 - val_mae: 0.4387
 Epoch 35/500
 295/295 34s 116ms/step -
 loss: 0.4379 - mae: 0.4379 - val_loss: 0.4431 - val_mae: 0.4431
 Epoch 36/500
 295/295 32s 109ms/step -
 loss: 0.4361 - mae: 0.4361 - val_loss: 0.4667 - val_mae: 0.4667
 Epoch 37/500
 295/295 32s 108ms/step -
 loss: 0.4305 - mae: 0.4305 - val_loss: 0.4355 - val_mae: 0.4355
 Epoch 38/500

295/295 33s 110ms/step -
 loss: 0.4295 - mae: 0.4295 - val_loss: 0.4250 - val_mae: 0.4250
 Epoch 39/500
 295/295 32s 108ms/step -
 loss: 0.4298 - mae: 0.4298 - val_loss: 0.4628 - val_mae: 0.4628
 Epoch 40/500
 295/295 32s 109ms/step -
 loss: 0.4271 - mae: 0.4271 - val_loss: 0.4377 - val_mae: 0.4377
 Epoch 41/500
 295/295 32s 107ms/step -
 loss: 0.4251 - mae: 0.4251 - val_loss: 0.4229 - val_mae: 0.4229
 Epoch 42/500
 295/295 31s 106ms/step -
 loss: 0.4187 - mae: 0.4187 - val_loss: 0.4252 - val_mae: 0.4252
 Epoch 43/500
 295/295 32s 108ms/step -
 loss: 0.4200 - mae: 0.4200 - val_loss: 0.4206 - val_mae: 0.4206
 Epoch 44/500
 295/295 32s 108ms/step -
 loss: 0.4232 - mae: 0.4232 - val_loss: 0.4212 - val_mae: 0.4212
 Epoch 45/500
 295/295 32s 109ms/step -
 loss: 0.4138 - mae: 0.4138 - val_loss: 0.4215 - val_mae: 0.4215
 Epoch 46/500
 295/295 32s 109ms/step -
 loss: 0.4134 - mae: 0.4134 - val_loss: 0.4218 - val_mae: 0.4218
 Epoch 47/500
 295/295 31s 106ms/step -
 loss: 0.4105 - mae: 0.4105 - val_loss: 0.4178 - val_mae: 0.4178
 Epoch 48/500
 295/295 32s 108ms/step -
 loss: 0.4112 - mae: 0.4112 - val_loss: 0.4248 - val_mae: 0.4248
 Epoch 49/500
 295/295 32s 108ms/step -
 loss: 0.4112 - mae: 0.4112 - val_loss: 0.4111 - val_mae: 0.4111
 Epoch 50/500
 295/295 31s 104ms/step -
 loss: 0.4045 - mae: 0.4045 - val_loss: 0.4124 - val_mae: 0.4124
 Epoch 51/500
 295/295 32s 108ms/step -
 loss: 0.4060 - mae: 0.4060 - val_loss: 0.4067 - val_mae: 0.4067
 Epoch 52/500
 295/295 31s 105ms/step -
 loss: 0.4054 - mae: 0.4054 - val_loss: 0.4422 - val_mae: 0.4422
 Epoch 53/500
 295/295 32s 108ms/step -
 loss: 0.4093 - mae: 0.4093 - val_loss: 0.4009 - val_mae: 0.4009
 Epoch 54/500

295/295 32s 108ms/step -
 loss: 0.4005 - mae: 0.4005 - val_loss: 0.4159 - val_mae: 0.4159
 Epoch 55/500
 295/295 31s 107ms/step -
 loss: 0.3991 - mae: 0.3991 - val_loss: 0.4197 - val_mae: 0.4197
 Epoch 56/500
 295/295 33s 110ms/step -
 loss: 0.4003 - mae: 0.4003 - val_loss: 0.4018 - val_mae: 0.4018
 Epoch 57/500
 295/295 31s 106ms/step -
 loss: 0.3938 - mae: 0.3938 - val_loss: 0.3961 - val_mae: 0.3961
 Epoch 58/500
 295/295 32s 109ms/step -
 loss: 0.3910 - mae: 0.3910 - val_loss: 0.4274 - val_mae: 0.4274
 Epoch 59/500
 295/295 32s 109ms/step -
 loss: 0.3922 - mae: 0.3922 - val_loss: 0.3946 - val_mae: 0.3946
 Epoch 60/500
 295/295 33s 111ms/step -
 loss: 0.3852 - mae: 0.3852 - val_loss: 0.3943 - val_mae: 0.3943
 Epoch 61/500
 295/295 32s 108ms/step -
 loss: 0.3904 - mae: 0.3904 - val_loss: 0.4091 - val_mae: 0.4091
 Epoch 62/500
 295/295 33s 111ms/step -
 loss: 0.3903 - mae: 0.3903 - val_loss: 0.3896 - val_mae: 0.3896
 Epoch 63/500
 295/295 33s 113ms/step -
 loss: 0.3873 - mae: 0.3873 - val_loss: 0.4283 - val_mae: 0.4283
 Epoch 64/500
 295/295 32s 108ms/step -
 loss: 0.3845 - mae: 0.3845 - val_loss: 0.3874 - val_mae: 0.3874
 Epoch 65/500
 295/295 33s 110ms/step -
 loss: 0.3925 - mae: 0.3925 - val_loss: 0.3859 - val_mae: 0.3859
 Epoch 66/500
 295/295 32s 109ms/step -
 loss: 0.3824 - mae: 0.3824 - val_loss: 0.3928 - val_mae: 0.3928
 Epoch 67/500
 295/295 32s 110ms/step -
 loss: 0.3779 - mae: 0.3779 - val_loss: 0.3868 - val_mae: 0.3868
 Epoch 68/500
 295/295 32s 109ms/step -
 loss: 0.3799 - mae: 0.3799 - val_loss: 0.3900 - val_mae: 0.3900
 Epoch 69/500
 295/295 32s 107ms/step -
 loss: 0.3772 - mae: 0.3772 - val_loss: 0.4043 - val_mae: 0.4043
 Epoch 70/500

```

295/295          32s 109ms/step -
loss: 0.3748 - mae: 0.3748 - val_loss: 0.3838 - val_mae: 0.3838
Epoch 71/500
295/295          34s 114ms/step -
loss: 0.3755 - mae: 0.3755 - val_loss: 0.3952 - val_mae: 0.3952
Epoch 72/500
295/295          33s 112ms/step -
loss: 0.3761 - mae: 0.3761 - val_loss: 0.3741 - val_mae: 0.3741
Epoch 73/500
295/295          33s 110ms/step -
loss: 0.3721 - mae: 0.3721 - val_loss: 0.3775 - val_mae: 0.3775
Epoch 74/500
295/295          32s 108ms/step -
loss: 0.3725 - mae: 0.3725 - val_loss: 0.3744 - val_mae: 0.3744
Epoch 75/500
295/295          33s 111ms/step -
loss: 0.3706 - mae: 0.3706 - val_loss: 0.3768 - val_mae: 0.3768
Epoch 76/500
295/295          33s 112ms/step -
loss: 0.3712 - mae: 0.3712 - val_loss: 0.3771 - val_mae: 0.3771
Epoch 77/500
295/295          33s 112ms/step -
loss: 0.3687 - mae: 0.3687 - val_loss: 0.3741 - val_mae: 0.3741

```

23 EVALUATION

```

[16]: # -----
# EVALUATION
# -----
val_loss, val_mae = model.evaluate(val, y_val, verbose=0)
print(f"\nValidation MAE: {val_mae:.5f}")

test_loss, test_mae = model.evaluate(test, y_test, verbose=0)
print(f"\nTest MAE: {test_mae:.5f}")

```

Validation MAE: 0.37413

Test MAE: 0.37406

24 PLOTS

```

[17]: # -----
# PLOT TRAINING HISTORY
# -----
print('='*25, 'PLOT TRAINING HISTORY', '='*25)

```

```

plt.figure()
plt.plot(history.history['loss'])
plt.plot(history.history['val_loss'])
plt.title("Training vs Validation MAE")
plt.xlabel("Epoch")
plt.ylabel("MAE")
plt.legend(["Train", "Validation"])
plt.show()

# -----
# PREDICTIONS on validation set
# -----
y_pred = model.predict(val)
# -----
# Plots of 10 single breath
# -----
print('='*25, 'PLOT BREATH SAMPLES ON VALIDATION', '='*25)
num_samples = 10
indices = np.random.choice(len(y_val), num_samples, replace=False)

plt.figure(figsize=(15, 18))

for i, idx in enumerate(indices):
    plt.subplot(5, 2, i + 1)
    plt.plot(y_val[idx].flatten())
    plt.plot(y_pred[idx].flatten())
    plt.title(f"Breath {idx}")
    plt.xlabel("Timestep")
    plt.ylabel("Pressure")
    plt.legend(["True", "Predicted"])

plt.tight_layout()
plt.show()

# PREDICTIONS on validation set
# -----
print('='*25, 'PLOT BREATH SAMPLES ON TEST', '='*25)
y_pred = model.predict(test)
# -----
# Plots of 10 single breath
# -----
num_samples = 10
indices = np.random.choice(len(y_test), num_samples, replace=False)

plt.figure(figsize=(15, 18))

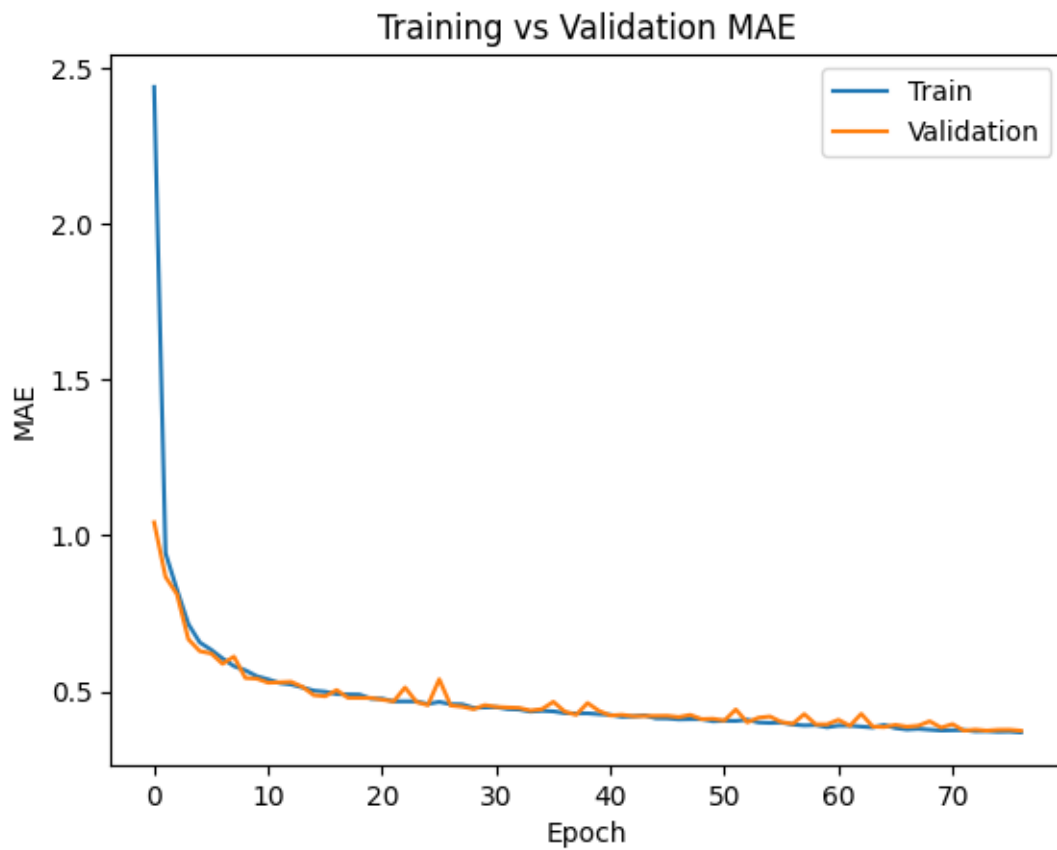
```

```

for i, idx in enumerate(indices):
    plt.subplot(5, 2, i + 1)
    plt.plot(y_test[idx].flatten())
    plt.plot(y_pred[idx].flatten())
    plt.title(f"Breath {idx}")
    plt.xlabel("Timestep")
    plt.ylabel("Pressure")
    plt.legend(["True", "Predicted"])
plt.tight_layout()
plt.show()

```

===== PLOT TRAINING HISTORY =====

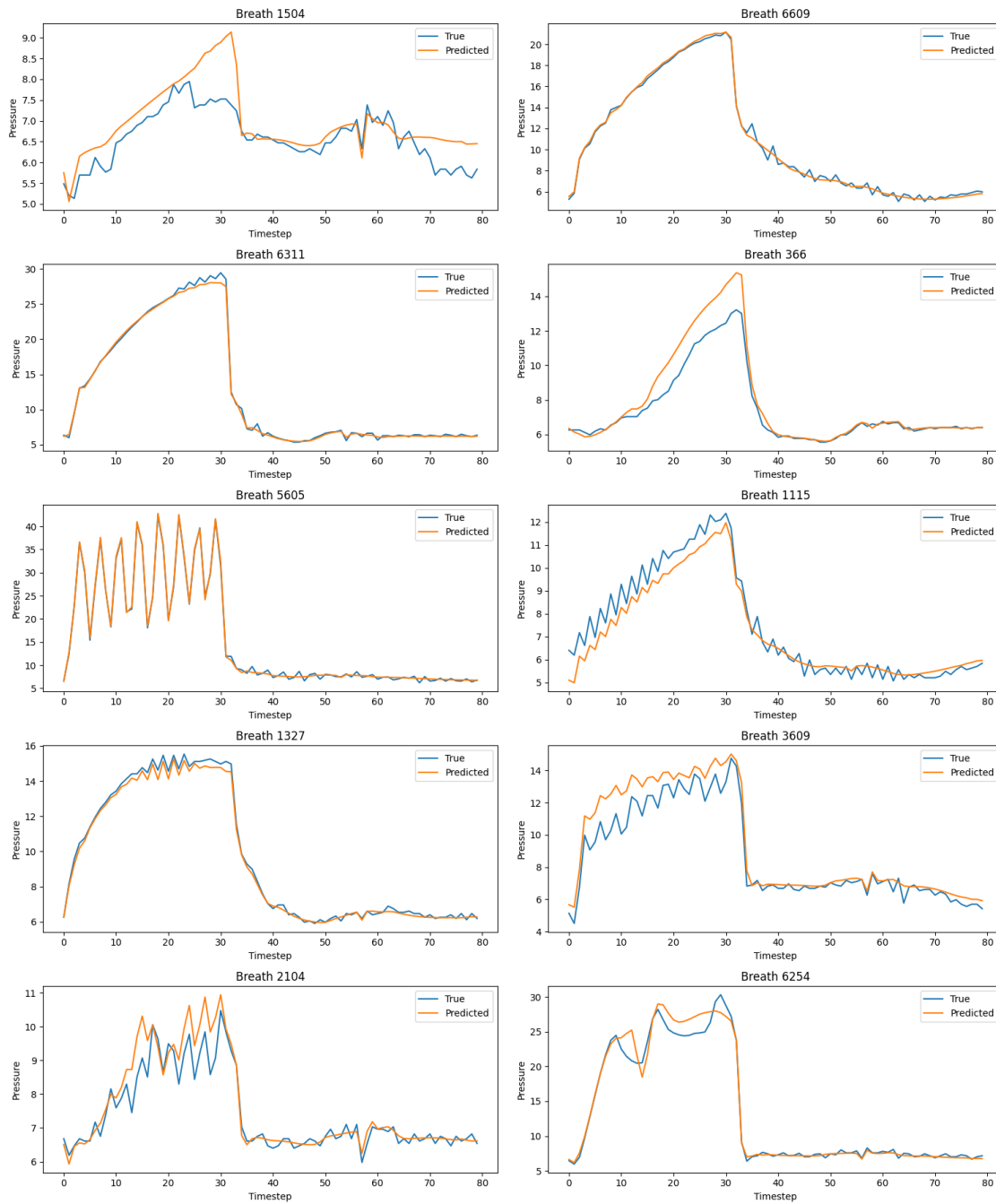


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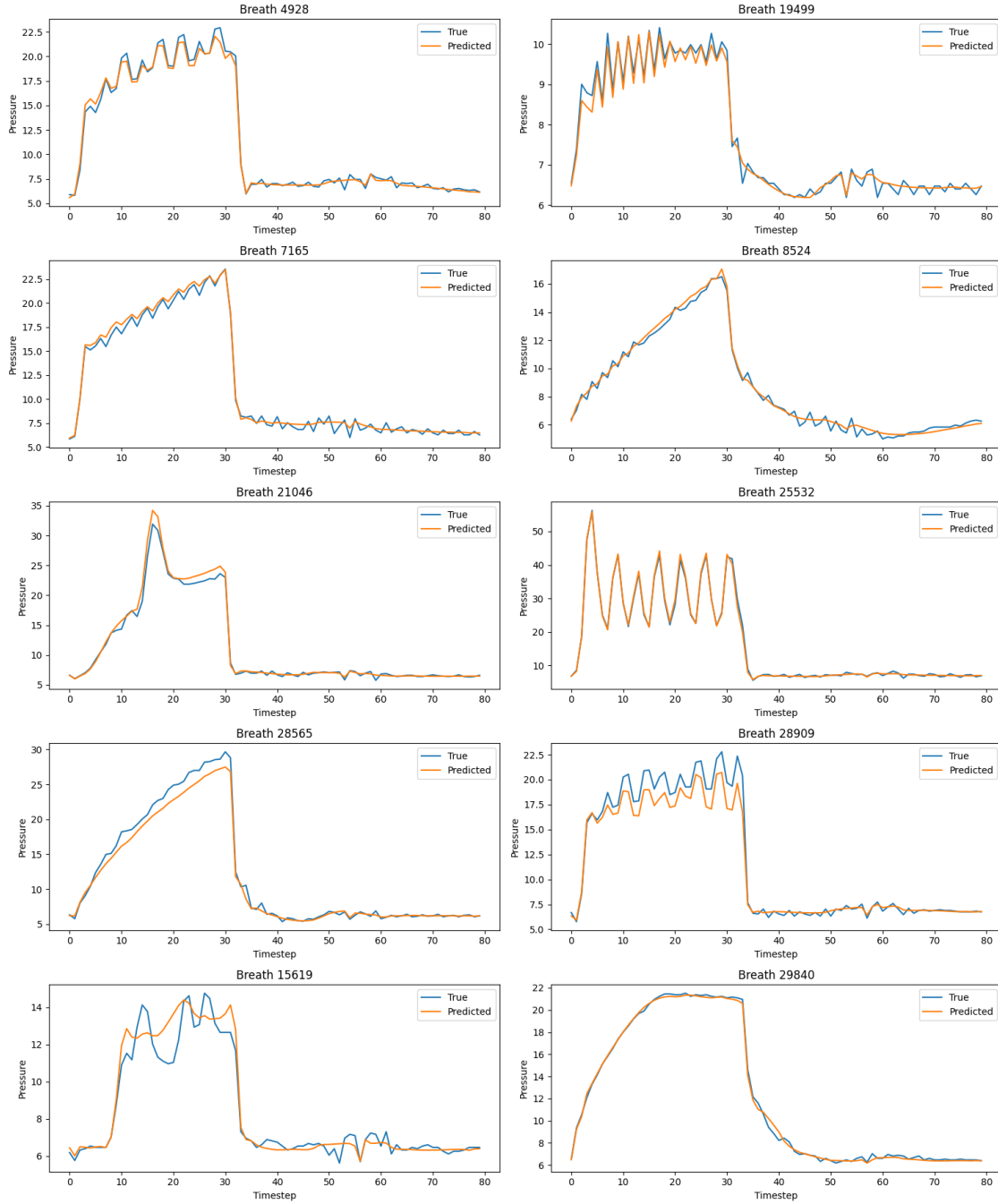
11s 39ms/step

===== PLOT BREATH SAMPLES ON VALIDATION

=====



===== PLOT BREATH SAMPLES ON TEST =====
 944/944 24s 26ms/step



25 Retrain on all train set

- We have to rebuild the model and train with whole train dataset to a number of epochs close to the early stopping value

```

[19]: # -----
# MODEL
# -----
input_shape = train.shape[2]
model = Sequential()

model.add(Bidirectional(
    LSTM(80, return_sequences=True),
    input_shape=(80, input_shape)
))

model.add(Bidirectional(LSTM(80, return_sequences=True)))
model.add(Bidirectional(LSTM(80, return_sequences=True)))

model.add(TimeDistributed(Dense(80)))
model.add(LeakyReLU(negative_slope=0.1))

model.add(TimeDistributed(Dense(80)))
model.add(LeakyReLU(negative_slope=0.1))

model.add(TimeDistributed(Dense(1)))

model.add(Flatten())

# -----
# COMPILE nd set optimizer
# -----
batch_size = 128
epochs = 80
steps_per_epoch = len(train) // batch_size
total_steps = steps_per_epoch * epochs

lr_schedule = tf.keras.optimizers.schedules.CosineDecay(
    initial_learning_rate=3e-4,
    decay_steps=total_steps,
    alpha=0.01
)

optimizer = tf.keras.optimizers.AdamW(
    learning_rate=lr_schedule,
    weight_decay=1e-4
)

model.compile(
    optimizer=optimizer,
    loss='mae',
    metrics=['mae']

```

```

)

model.summary()

# -----
# CALLBACKS
# -----
early_stop = EarlyStopping(
    monitor='val_loss',
    patience=5,
    restore_best_weights=True
)

# -----
# Train
# -----
# Make new train set
X_full = np.concatenate([train, val], axis=0)
y_full = np.concatenate([y_train, y_val], axis=0)

history = model.fit(
    X_full,
    y_full,
    epochs=epochs,
    batch_size=batch_size,
    verbose=1
)

# -----
# Save the model
# -----
model.save('/ventilator_retrained.keras')

```

Model: "sequential_2"

Layer (type)	Output Shape	Param #
bidirectional_6 (Bidirectional)	(None, 80, 160)	74,240
bidirectional_7 (Bidirectional)	(None, 80, 160)	154,240
bidirectional_8 (Bidirectional)	(None, 80, 160)	154,240
time_distributed_6 (TimeDistributed)	(None, 80, 80)	12,880

leaky_re_lu_4 (LeakyReLU)	(None, 80, 80)	0
time_distributed_7 (TimeDistributed)	(None, 80, 80)	6,480
leaky_re_lu_5 (LeakyReLU)	(None, 80, 80)	0
time_distributed_8 (TimeDistributed)	(None, 80, 1)	81
flatten_2 (Flatten)	(None, 80)	0

Total params: 402,161 (1.53 MB)

Trainable params: 402,161 (1.53 MB)

Non-trainable params: 0 (0.00 B)

Epoch 1/80

354/354 46s 102ms/step -

loss: 2.0339 - mae: 2.0339

Epoch 2/80

354/354 36s 101ms/step -

loss: 0.8955 - mae: 0.8955

Epoch 3/80

354/354 36s 102ms/step -

loss: 0.7511 - mae: 0.7511

Epoch 4/80

354/354 35s 99ms/step -

loss: 0.6747 - mae: 0.6747

Epoch 5/80

354/354 37s 104ms/step -

loss: 0.6394 - mae: 0.6394

Epoch 6/80

354/354 36s 103ms/step -

loss: 0.6107 - mae: 0.6107

Epoch 7/80

354/354 37s 103ms/step -

loss: 0.5856 - mae: 0.5856

Epoch 8/80

354/354 38s 106ms/step -

loss: 0.5626 - mae: 0.5626

Epoch 9/80

354/354 37s 104ms/step -

loss: 0.5476 - mae: 0.5476

Epoch 10/80
 354/354 36s 102ms/step -
 loss: 0.5297 - mae: 0.5297
 Epoch 11/80
 354/354 38s 107ms/step -
 loss: 0.5190 - mae: 0.5190
 Epoch 12/80
 354/354 37s 105ms/step -
 loss: 0.5121 - mae: 0.5121
 Epoch 13/80
 354/354 37s 104ms/step -
 loss: 0.5010 - mae: 0.5010
 Epoch 14/80
 354/354 38s 108ms/step -
 loss: 0.4922 - mae: 0.4922
 Epoch 15/80
 354/354 37s 105ms/step -
 loss: 0.4840 - mae: 0.4840
 Epoch 16/80
 354/354 37s 106ms/step -
 loss: 0.4791 - mae: 0.4791
 Epoch 17/80
 354/354 37s 104ms/step -
 loss: 0.4746 - mae: 0.4746
 Epoch 18/80
 354/354 36s 103ms/step -
 loss: 0.4735 - mae: 0.4735
 Epoch 19/80
 354/354 36s 103ms/step -
 loss: 0.4622 - mae: 0.4622
 Epoch 20/80
 354/354 37s 106ms/step -
 loss: 0.4597 - mae: 0.4597
 Epoch 21/80
 354/354 37s 106ms/step -
 loss: 0.4566 - mae: 0.4566
 Epoch 22/80
 354/354 37s 104ms/step -
 loss: 0.4513 - mae: 0.4513
 Epoch 23/80
 354/354 37s 105ms/step -
 loss: 0.4477 - mae: 0.4477
 Epoch 24/80
 354/354 38s 106ms/step -
 loss: 0.4478 - mae: 0.4478
 Epoch 25/80
 354/354 37s 106ms/step -
 loss: 0.4402 - mae: 0.4402

Epoch 26/80
 354/354 36s 103ms/step -
 loss: 0.4359 - mae: 0.4359
 Epoch 27/80
 354/354 37s 105ms/step -
 loss: 0.4325 - mae: 0.4325
 Epoch 28/80
 354/354 37s 104ms/step -
 loss: 0.4313 - mae: 0.4313
 Epoch 29/80
 354/354 37s 105ms/step -
 loss: 0.4254 - mae: 0.4254
 Epoch 30/80
 354/354 37s 104ms/step -
 loss: 0.4274 - mae: 0.4274
 Epoch 31/80
 354/354 37s 105ms/step -
 loss: 0.4220 - mae: 0.4220
 Epoch 32/80
 354/354 36s 103ms/step -
 loss: 0.4171 - mae: 0.4171
 Epoch 33/80
 354/354 37s 106ms/step -
 loss: 0.4165 - mae: 0.4165
 Epoch 34/80
 354/354 37s 106ms/step -
 loss: 0.4134 - mae: 0.4134
 Epoch 35/80
 354/354 37s 104ms/step -
 loss: 0.4113 - mae: 0.4113
 Epoch 36/80
 354/354 38s 106ms/step -
 loss: 0.4096 - mae: 0.4096
 Epoch 37/80
 354/354 37s 104ms/step -
 loss: 0.4060 - mae: 0.4060
 Epoch 38/80
 354/354 38s 107ms/step -
 loss: 0.4051 - mae: 0.4051
 Epoch 39/80
 354/354 38s 107ms/step -
 loss: 0.4016 - mae: 0.4016
 Epoch 40/80
 354/354 39s 110ms/step -
 loss: 0.3995 - mae: 0.3995
 Epoch 41/80
 354/354 36s 102ms/step -
 loss: 0.3959 - mae: 0.3959

Epoch 42/80
 354/354 36s 102ms/step -
 loss: 0.3949 - mae: 0.3949
 Epoch 43/80
 354/354 37s 104ms/step -
 loss: 0.3932 - mae: 0.3932
 Epoch 44/80
 354/354 37s 105ms/step -
 loss: 0.3909 - mae: 0.3909
 Epoch 45/80
 354/354 37s 105ms/step -
 loss: 0.3892 - mae: 0.3892
 Epoch 46/80
 354/354 38s 107ms/step -
 loss: 0.3877 - mae: 0.3877
 Epoch 47/80
 354/354 38s 109ms/step -
 loss: 0.3858 - mae: 0.3858
 Epoch 48/80
 354/354 38s 106ms/step -
 loss: 0.3842 - mae: 0.3842
 Epoch 49/80
 354/354 37s 105ms/step -
 loss: 0.3818 - mae: 0.3818
 Epoch 50/80
 354/354 38s 106ms/step -
 loss: 0.3802 - mae: 0.3802
 Epoch 51/80
 354/354 37s 105ms/step -
 loss: 0.3789 - mae: 0.3789
 Epoch 52/80
 354/354 38s 107ms/step -
 loss: 0.3786 - mae: 0.3786
 Epoch 53/80
 354/354 39s 109ms/step -
 loss: 0.3776 - mae: 0.3776
 Epoch 54/80
 354/354 38s 108ms/step -
 loss: 0.3756 - mae: 0.3756
 Epoch 55/80
 354/354 37s 104ms/step -
 loss: 0.3747 - mae: 0.3747
 Epoch 56/80
 354/354 38s 107ms/step -
 loss: 0.3729 - mae: 0.3729
 Epoch 57/80
 354/354 38s 106ms/step -
 loss: 0.3722 - mae: 0.3722

Epoch 58/80
354/354 38s 108ms/step -
loss: 0.3712 - mae: 0.3712
Epoch 59/80
354/354 37s 104ms/step -
loss: 0.3705 - mae: 0.3705
Epoch 60/80
354/354 39s 110ms/step -
loss: 0.3699 - mae: 0.3699
Epoch 61/80
354/354 38s 107ms/step -
loss: 0.3693 - mae: 0.3693
Epoch 62/80
354/354 38s 107ms/step -
loss: 0.3688 - mae: 0.3688
Epoch 63/80
354/354 38s 107ms/step -
loss: 0.3685 - mae: 0.3685
Epoch 64/80
354/354 38s 108ms/step -
loss: 0.3681 - mae: 0.3681
Epoch 65/80
354/354 37s 104ms/step -
loss: 0.3679 - mae: 0.3679
Epoch 66/80
354/354 38s 108ms/step -
loss: 0.3676 - mae: 0.3676
Epoch 67/80
354/354 38s 106ms/step -
loss: 0.3676 - mae: 0.3676
Epoch 68/80
354/354 38s 107ms/step -
loss: 0.3676 - mae: 0.3676
Epoch 69/80
354/354 37s 105ms/step -
loss: 0.3675 - mae: 0.3675
Epoch 70/80
354/354 38s 108ms/step -
loss: 0.3674 - mae: 0.3674
Epoch 71/80
354/354 38s 108ms/step -
loss: 0.3673 - mae: 0.3673
Epoch 72/80
354/354 38s 108ms/step -
loss: 0.3673 - mae: 0.3673
Epoch 73/80
354/354 37s 104ms/step -
loss: 0.3673 - mae: 0.3673

```

Epoch 74/80
354/354          38s 108ms/step -
loss: 0.3672 - mae: 0.3672
Epoch 75/80
354/354          37s 106ms/step -
loss: 0.3671 - mae: 0.3671
Epoch 76/80
354/354          38s 108ms/step -
loss: 0.3670 - mae: 0.3670
Epoch 77/80
354/354          38s 107ms/step -
loss: 0.3670 - mae: 0.3670
Epoch 78/80
354/354          38s 107ms/step -
loss: 0.3670 - mae: 0.3670
Epoch 79/80
354/354          38s 108ms/step -
loss: 0.3669 - mae: 0.3669
Epoch 80/80
354/354          38s 106ms/step -
loss: 0.3669 - mae: 0.3669

```

26 Plots

```

[26]: # PREDICTIONS on validation set
# -----
print('='*25, 'PLOT BREATH SAMPLES ON TEST', '='*25)
y_pred = model.predict(test)
# -----
# Plots of 10 single breath
# -----
num_samples = 10
indices = np.random.choice(len(y_test), num_samples, replace=False)

plt.figure(figsize=(15, 18))

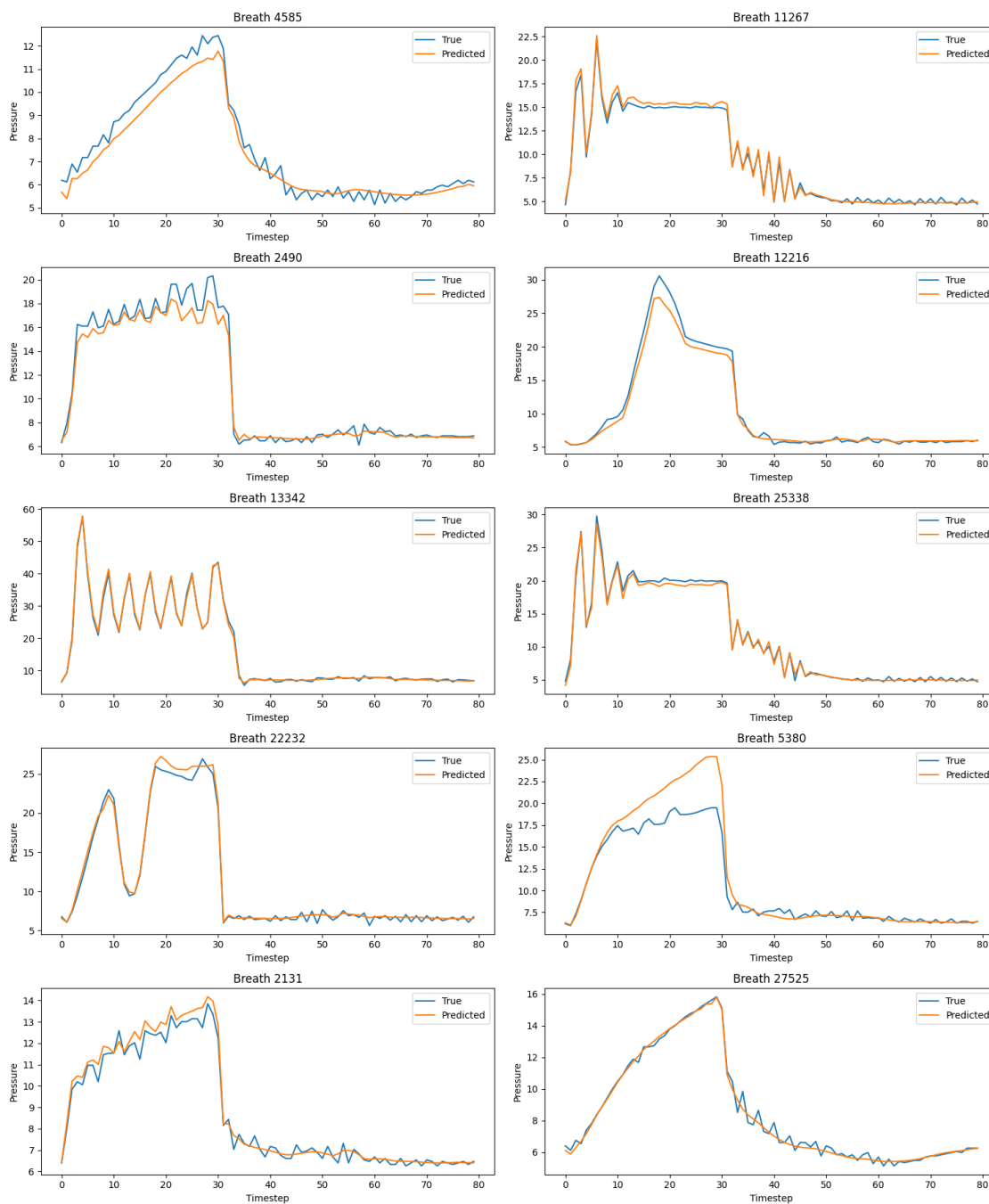
for i, idx in enumerate(indices):
    plt.subplot(5, 2, i + 1)
    plt.plot(y_test[idx].flatten())
    plt.plot(y_pred[idx].flatten())
    plt.title(f"Breath {idx}")
    plt.xlabel("Timestep")
    plt.ylabel("Pressure")
    plt.legend(["True", "Predicted"])
plt.tight_layout()
plt.show()

```

===== PLOT BREATH SAMPLES ON TEST =====

944/944

26s 27ms/step



27 SHAP

```
[ ]: # -----  
# Initializes the JavaScript visualization environment used by the SHAP library  
# to display interactive plots in Jupyter notebooks.  
# -----  
shap.initjs()
```

<IPython.core.display.HTML object>

```
[ ]: # -----  
# Extracting time-series dimensions  
# -----  
n_timesteps = train.shape[1]  
n_features = train.shape[2]  
  
# -----  
# Flattening the time-series  
# -----  
train_flat = train.reshape(train.shape[0], -1)  
test_flat = test.reshape(test.shape[0], -1)  
  
# -----  
# Define a prediction wrapper  
# -----  
def predict_fn(x):  
    x = x.reshape(x.shape[0], n_timesteps, n_features)  
    return model.predict(x)  
  
print(train_flat.shape)
```

(37725, 2800)

```
[ ]: # -----  
# Selects the first 10 samples from the flattened training set.  
# -----  
background = train_flat[:10]  
  
# -----  
# Creating the Kernel SHAP explainer  
# -----  
explainer = shap.KernelExplainer(predict_fn, background)  
  
# -----  
# Computing SHAP values  
# -----  
shap_values = explainer.shap_values(test_flat[:5])
```

```
print(shap_values.shape)
```

```
1/1          1s 1s/step
 0%|          | 0/5 [00:00<?, ?it/s]

1/1          2s 2s/step
1632/1632      43s 26ms/step
 20%|          | 1/5 [01:16<05:06, 76.59s/it]

1/1          0s 75ms/step
1732/1732      45s 26ms/step
 40%|          | 2/5 [02:28<03:41, 73.75s/it]

1/1          0s 76ms/step
1632/1632      42s 26ms/step
 60%|          | 3/5 [03:34<02:20, 70.09s/it]

1/1          0s 78ms/step
1632/1632      43s 27ms/step
 80%|          | 4/5 [04:40<01:08, 68.80s/it]

1/1          0s 78ms/step
1632/1632      44s 27ms/step
100%|          | 5/5 [05:49<00:00, 69.94s/it]
```

```
[ ]: # -----
# Convert to array
# -----
shap_values_array = np.array(shap_values) # (5, 2800, 80)

# -----
# Reshape input features back to (timesteps, features)
# -----
shap_values_array = shap_values_array.reshape(
    test_flat[:5].shape[0], # samples
    n_timesteps,           # input timesteps
    n_features,            # features
    n_timesteps            # output timesteps
)
# (samples, input_timesteps, features, output_timesteps)
# i.e., (5, 80, 35, 80)

# -----
# Take the diagonal (own timestep features)
# -----
shap_diag = shap_values_array[np.arange(5)[: , None], np.arange(n_timesteps), :,
↪ np.arange(n_timesteps)]
```

```
# shape: (samples, timesteps, features) -> (5, 80, 35)

print("SHAP diagonal shape:", shap_diag.shape)
# SHAP diagonal shape: (5, 80, 35)
```

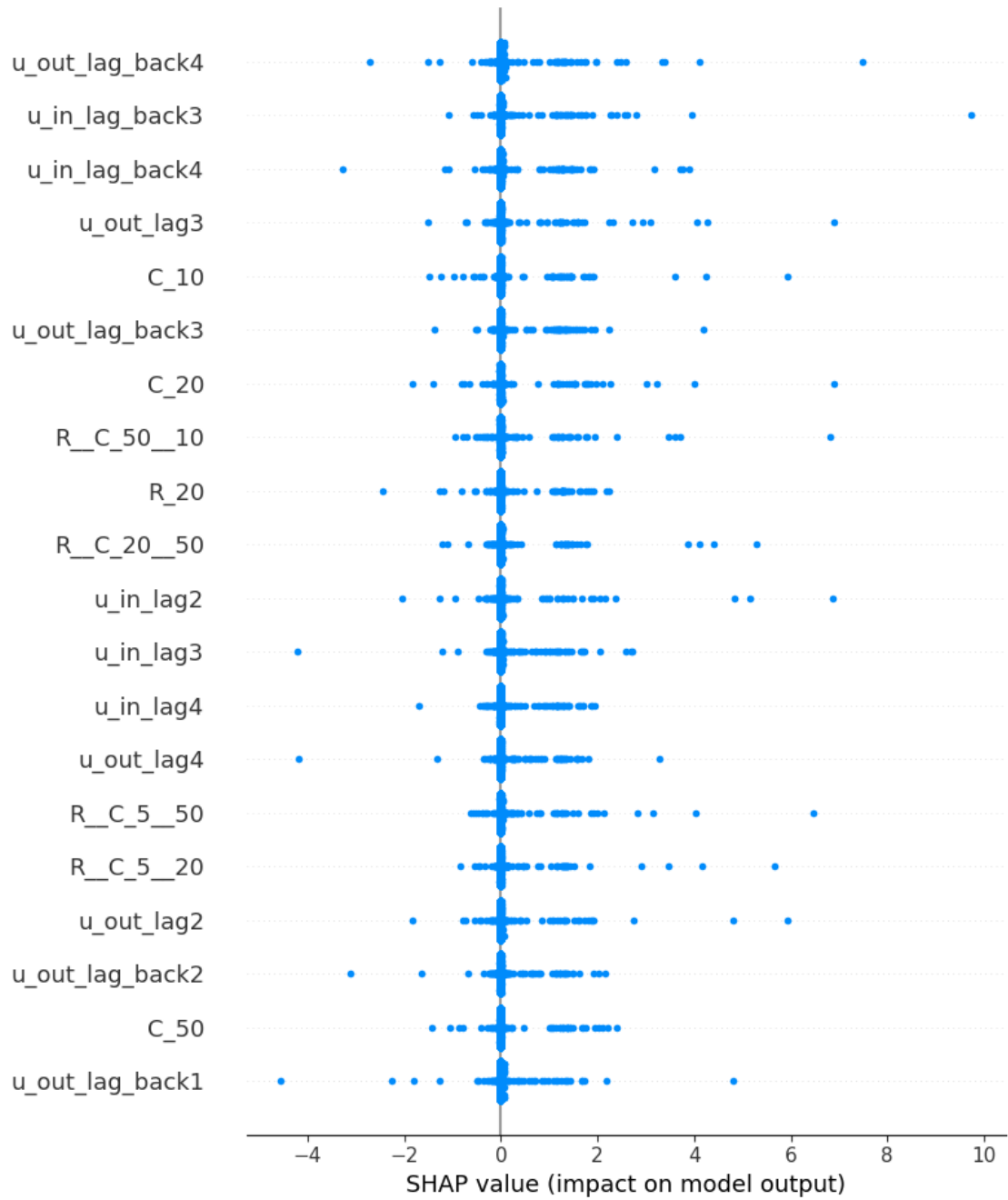
27.1 Shap Plots

What This Plot Represents

- Y-axis: features used by the model
- X-axis: SHAP value = impact on prediction
- Each dot: one sample from the dataset
- Distance from 0: strength of the feature's influence

```
[ ]: # flatten for summary plot: (samples*timesteps, features)
shap_flat = shap_diag.reshape(-1, n_features)
X_flat = test[:5].reshape(-1, n_features)

shap.summary_plot(
    shap_flat,
    X_flat,
    feature_names=features
)
```



Interpretation:

- $\text{SHAP} > 0 \rightarrow$ pushes prediction higher
- $\text{SHAP} < 0 \rightarrow$ pushes prediction lower
- More spread $\rightarrow$ more important feature

Top contributors appear to be:

- u_out_lag1
- u_in_lag3
- u_out_lag_back2
- u_out_lag_back1
- u_in_lag_back1
- u_in_lag2
- u_out_lag2

This tells us your LSTM heavily depends on recent past values.

This is exactly what we expect in time-series dynamics.

Medium Importance Features

These still influence predictions but less strongly:

- R__C__20__50
- R__C__50__50
- R__C__5__50
- R__C__50__10
- R__C__5__10
- area

These are system configuration parameters, meaning the model uses them but dynamics dominate.

Lowest Importance

- u_in_cumsum

This feature seems to contribute the least.

Possible reasons:

redundant with lag features

LSTM already captures cumulative information internally

One Important Limitation of This Plot

This SHAP plot collapses the time dimension.

SHAP summarizes it only per feature, not per time step.

A more informative plot would be:

feature $\times$ timestep importance

which shows **when** each feature matters.

28 SHAP Temporal Importance Heatmap

28.1 Plot heatmap

This plot is a temporal SHAP heatmap, which shows how each feature influences the prediction at each timestep of our sequence.

X-axis : Timestep (0–79) Each column represents a time position in the sequence.

Our model uses 80 timesteps.

Y-axis : Features - Examples:

- u_in

- u_out
- lag features (u_in_lag1, u_out_lag2, etc.)
- system parameters (R, C)
- engineered features (area, u_in_cumsum)

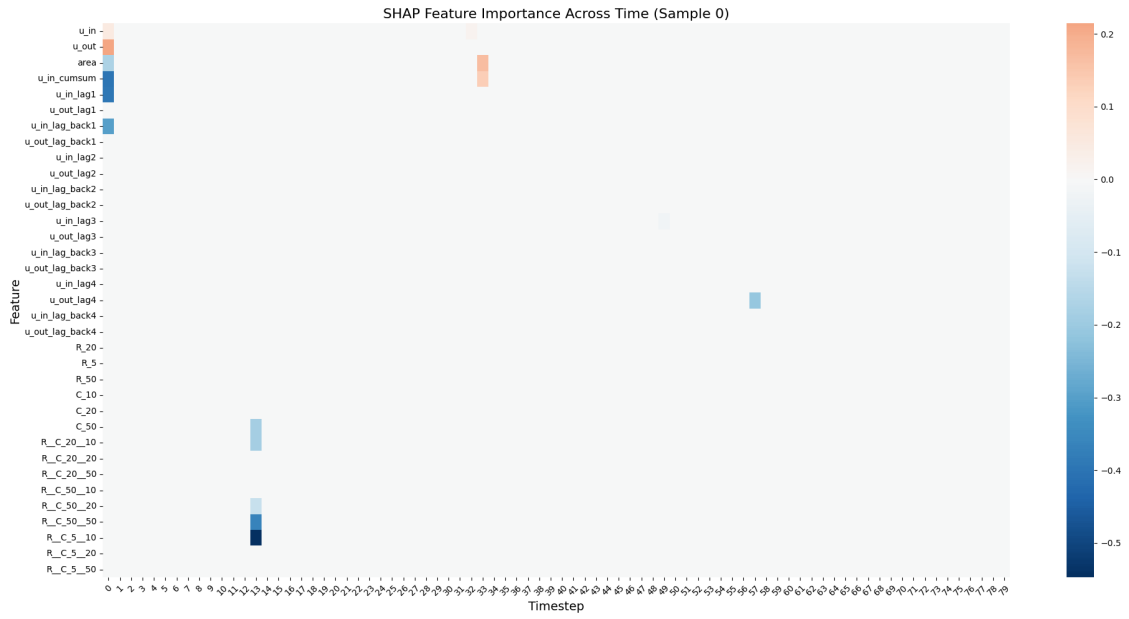
Color Meanings:

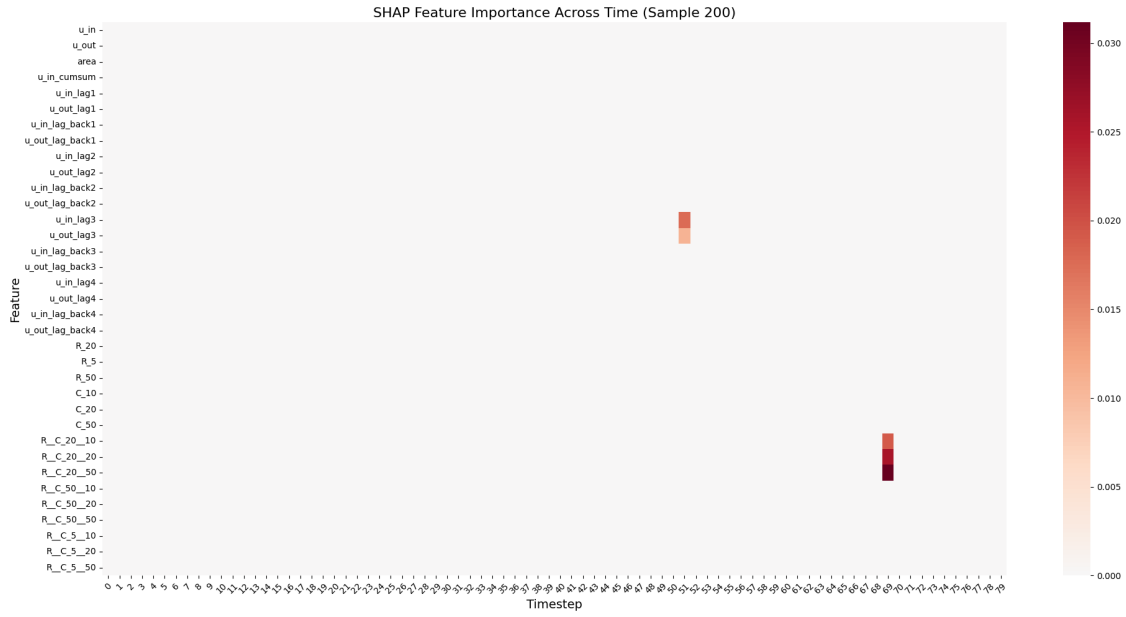
Color	Meaning
Red	Feature increases prediction
Blue	Feature decreases prediction
White	Feature little or no effect

```
[ ]: for instance in range(shap_diag.shape[0]):
    plt.figure(figsize=(20, 10)) # width x height in inches
    sns.heatmap(
        shap_diag[instance].T,
        xticklabels=range(n_timesteps),
        yticklabels=features,
        cmap="RdBu_r",
        center=0,
        vmax = np.abs(shap_diag).max(),
        vmin = - np.abs(shap_diag).max(),
    )

    plt.xlabel("Timestep", fontsize=14)
    plt.ylabel("Feature", fontsize=14)
    plt.title(f"SHAP Feature Importance Across Time (Sample {instance})",
    ↪ fontsize=16)

    plt.xticks(rotation=45) # rotate x-axis labels for readability
    plt.yticks(rotation=0) # keep y-axis labels horizontal
    plt.tight_layout() # adjust layout to prevent cut-offs
    plt.show()
```





29 Important Patterns in our Plot

Important timesteps appears to be: $t = 24$

$t = 33$

$t = 50$

$t = 78$

Lag Features Are Dominant.

System Parameters Affect Prediction Later.

Early Timesteps Matter Less.

We can notice how most of the left side of the plot is empty. That means:

$t = 0-15$

has almost no influence on predictions.

At timestep ~ 50 : many features turn blue

which implies that they push predictions down

That suggests a phase transition in the breathing cycle.

Caveat: SHAP plot shows local explanation, not global importance. To see global feature-time importance we should run SHAP across all samples which is a task very consuming time.